

Coordinate geometry

Gradients, distances, midpoints and straight lines

YOUR AIM TODAY

Build straight-line geometry from the coordinate basics.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

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Start time: _____ Finish time: _____

Date: _____

The ideas you need

A point is written (x, y) . A line is controlled by its gradient and one point.

- Gradient $m = (\text{change in } y) / (\text{change in } x)$.
- Midpoint of (x_1, y_1) and (x_2, y_2) is $((x_1 + x_2)/2, (y_1 + y_2)/2)$.
- Distance is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- Line through (x_1, y_1) : $y - y_1 = m(x - x_1)$.
- Parallel lines have equal gradients; perpendicular gradients multiply to -1 .

MEMORY RULE

Find the gradient first, then substitute one known point.

Line through two points

WORKED STEPS

Through $(2, 3)$ and $(6, 11)$: $m = (11 - 3) / (6 - 2) = 2$. So $y - 3 = 2(x - 2)$, hence $y = 2x - 1$.

Perpendicular line

WORKED STEPS

A line perpendicular to $y = 3x + 4$ has gradient $-1/3$.

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Find the gradient from (1,2) to (5,10).	
2. Find the midpoint of (-2,5) and (6,-1).	
3. Find the distance between (1,1) and (4,5).	
4. Find the equation through (3,2) with gradient 4.	
5. Find the equation through (1,5) and (3,1).	
6. Find the perpendicular bisector of (0,0) and (4,2).	

STOP

Do not continue until all six questions have an attempt.

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	$m=(10-2)/(5-1)=2.$
2	$(2,2).$
3	$\text{sqrt}(3^2+4^2)=5.$
4	$y-2=4(x-3), \text{ so } y=4x-10.$
5	$m=-2; y-5=-2(x-1), \text{ so } y=-2x+7.$
6	Midpoint $(2,1)$; segment gradient $1/2$, so perpendicular gradient -2 . $y-1=-2(x-2)$, so $y=-2x+5.$

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.